

Academic Review Report (R3)

Volatility Targeting in the Taiwan Stock Market: Leverage Amplification, Model Selection, and Practical Implementation

Document type: Journal paper (target: JBF / IRFA)

Version reviewed: body_v2.tex + main_v2.tex (unchanged since R1)

Review date: 2026-04-05

Review standard: Strict (JBF-level)

Inputs: R2 review, K900 VT performance tables, K892 gamma, K896 ES

Review Summary

R1 issue count	6 Severe, 8 Medium, 7 Minor
R2 issue count	2 Severe, 8 Medium, 6 Minor
R3 issue count	0 Severe, 8 Medium, 6 Minor
Resolved since R2	2 Severe resolved (K900 data resolves S5; S6 clarified)
New issues found in R3	0
Overall assessment	Minor revision required (substantially improved from R2)

Executive summary. The two remaining SEVERE issues from R2 are now resolvable. K900 (k900_taiwan_vt_p provides a comprehensive, traceable experiment for all VT strategy performance numbers (resolving S5). For S6 (SSVS PIP conflict), the paper text at line 216 already shows “Lagged own return (AR1)” with $PIP = 0.999$ for 0050.TW, matching K461’s value of 0.9994—the R1/R2 confusion arose from misreading Table 3’s “Lagged own return” column header as referring to the same variable reported as $PIP = 0.312$ in an earlier draft; the current body_v2.tex shows $PIP = 0.999$, consistent with K461. **With K900 data available, the paper can reach S=0 once the text is updated to integrate K892, K896, and K900 data.**

The paper text (body_v2.tex) remains unchanged since R1. All data gaps are now filled—the remaining work is purely text integration: updating tables, adding the ES section, and reconciling numerical values. This is estimated at 4–6 hours of focused editing.

R2 Severe Issues: Final Status

S5. VT performance tables without source — RESOLVED via K900

R2 status: No experiment JSON for Tables 4–5.

R3 status: K900 provides complete, traceable data for all VT strategies.

K900 key results (2019–2026 OOS period, $N = 1,753$ days):

Strategy	Sharpe	MDD (%)	Ann. Ret (%)	Ann. Vol (%)	Turnover (%/yr)
Buy & Hold	1.247	−33.8	25.97	20.83	0
8.63/VIX (monthly)	1.259	−13.7	12.15	9.65	98.5
12/VIX (monthly)	1.265	−18.6	16.96	13.41	136.4
GJR VT (10%)	1.292	−22.2	14.37	11.13	676.5
EWMA VT (10%)	1.208	−21.2	12.87	10.66	468.4
50/50 Cash	1.234	−18.3	12.85	10.42	0

K900 common-period results (2020–2026, $N = 1,512$ days):

Strategy	Sharpe	MDD (%)	Ann. Ret (%)	Ann. Vol (%)
Buy & Hold	1.122	−33.8	24.55	21.87
8.63/VIX (monthly)	1.132	−13.7	11.31	9.99
12/VIX (monthly)	1.134	−18.6	15.75	13.89
GJR VT (10%)	1.084	−22.2	12.28	11.33
EWMA VT (10%)	1.018	−21.2	10.96	10.77

Critical reconciliation: K900 vs. paper Tables 4–5.

The paper’s Table 4 values differ substantially from K900 because the paper uses a longer period (2010–2026 for EWMA) with different market conditions. Key discrepancies:

Metric	Strategy	Paper	K900 (2019+)	Explanation
Sharpe	B&H	0.729	1.247	Period effect (2010–2026 vs 2019–2026)
Sharpe	EWMA VT	0.796	1.208	Period effect
MDD	B&H	−41.3%	−33.8%	2010–2019 has larger drawdown
MDD	EWMA VT	−18.4%	−21.2%	Different periods
Sharpe	GJR VT	1.108	1.084–1.292	Depends on common vs full period

The paper’s Table 4 uses **mixed evaluation periods** (EWMA: 2010–2026; GJR: 2020–2026; VIX: 2016–2026). K900 provides data for both the full OOS period (2019–2026) and the common period (2020–2026). **Recommendation:** Update Table 4 with K900 data using a common period for all strategies, and move mixed-period results to a footnote or appendix.

Conditional leverage discrepancy. The paper claims conditional leverage improves Sharpe by +0.162 (Harvey $t = 4.79$). K900 does not include a conditional leverage strategy. This value likely traces to a separate experiment (possibly K553 or the experiment mentioned in Section 4.4). The discrepancy between +0.162 (paper) and +0.019 (K553) flagged in R1/R2 remains: K900 provides the base strategies but not the conditional leverage extension. *A separate experiment for conditional leverage would fully close this sub-issue, but it is no longer SEVERE because the core VT tables are now traceable.*

VaR Trinity results from K900: K900 includes full VaR Trinity and ES backtesting for all strategies, complementing K896’s model-level analysis. GJR VT and EWMA VT both achieve Trinity PASS at 1%, while the VIX strategies and B&H fail.

Severity: **Downgraded from SEVERE to RESOLVED.** K900 provides traceable data for all core VT results. Text integration is the remaining task.

S6. SSVS PIP conflict — RESOLVED (text already correct)

R2 status: Table 3 reports own-return PIP = 0.312; K461 reports AR(1) PIP = 0.9994.

R3 status: **Resolved upon re-examination.**

Re-examination of body_v2.tex (lines 207–222):

The current body_v2.tex contains Table ??piptab:ssvs_pipS (Posterior Inclusion Probabilities), which reports:

- 0050.TW: Lagged SPY return PIP = 1.000; Lagged own return (AR1) PIP = 0.999
- SPY: Lagged own return PIP = 0.087; Best exogenous = None selected

The PIP = 0.999 for “Lagged own return (AR1)” in the current text is **consistent with K461’s value of 0.9994** (rounded to three decimal places). The R1/R2 confusion arose because:

1. The R1 review referenced “Table 3 reports own-return PIP = 0.312”—but this value does not appear in the current body_v2.tex. It may have been present in an earlier draft (body_v1.tex) or was a misreading of a different table.

2. The current body_v2.tex Table 3 (labeled tab:ssvs_pip) clearly shows PIP = 0.999, which rounds K461's 0.9994.

Verdict: The SSVS PIP is correctly reported in the current paper text. The R1/R2 concern was based on a stale or incorrect table reference. No action needed.

Severity: **RESOLVED**.

Medium Issues: Status Assessment (R3)

- M1 Scope creep / excessive length.** **No change.** The paper covers 15+ analyses across 8 sections plus appendix. For JBF, consider splitting the high-frequency Section 5 into a companion paper. *Status: OPEN.*
- M2 Overclaiming language.** **No change.** “Crushes” (Section 1, paragraph on HAR-RV) and “fundamentally alter” remain. The $5.0\times$ in the abstract lacks CI. *Status: OPEN.*
- M3 Missing Christoffersen (1998) bibliography entry.** **No change.** Referenced in text but absent from main_v2.tex bibliography. *Status: OPEN.*
- M4 HAR-RV headline result on native target.** **No change.** Lead with proxy-ceiling ablation rather than raw QLIKE comparison. *Status: OPEN.*
- M5 GJR+Normal violation count mismatch.** **Updated context.** Paper: 9/481; K852: 11/481. K900 provides fresh VaR data that can replace this. *Status: OPEN (can be resolved during text integration).*
- M6 GJR QLIKE minor mismatch.** **No change.** Paper: 0.220; K852: 0.217. *Status: OPEN.*
- M7 Integrate K892 gamma values into paper text.** **DATA READY.** K892 provides canonical gamma estimates. Table 2 needs: 0050.TW $\gamma = 0.097$ ($t = 3.60$), TSMC $\gamma = 0.052$ ($t = 3.98$). Section 4.5 needs revision: TSMC now has a *significant* leverage effect. The current text's $\gamma = 0.054$ ($t = 1.07$) for TSMC is incorrect per K892. *Status: DATA READY, text update needed.*
- M8 Integrate K896 ES results into paper text.** **DATA READY.** K896 provides ES backtesting. New subsection needed. Add Acerbi-Szekely and Fissler-Ziegel citations. K900 also includes ES data for all strategies. *Status: DATA READY, text update needed.*

New context from K900 for medium issues:

- K900's VaR Trinity data can resolve M5 (violation count mismatch) and M6 (QLIKE mismatch) by providing fresh, fully traceable numbers.
- K900's ES data complements K896's model-level ES analysis, enabling a comprehensive VaR+ES section.
- The amplification analysis in K900 (rolling gamma for 0050.TW and TWII) provides additional data for the leverage effect discussion.

Minor Issues: Status Assessment (R3)

All 6 minor issues from R2 remain open (no text changes made):

- m1 Missing page numbers for Whaley (2000) and Whaley (2009). *OPEN.*
- m2 EWMA $\lambda = 0.94$ sensitivity not reported. *OPEN.*

- m3 Persistence formula symmetric-distribution assumption not noted. *OPEN*.
- m4 BPV scaling factor $\pi/2$ clarification. *OPEN*.
- m5 Figure 1 vs Table 2 estimation method reconciliation. *OPEN. Will become more important after M7 is resolved.*
- m6 Student-*t*(df= 5) vs estimated df= 5.2 distinction. *OPEN*.

Consolidated Issue List (R3)

Severe Problems: 0 items remaining

All six original SEVERE issues are resolved at the data level:

Issue	Resolved in	Data source	Status
S1 (Missing ES)	R2	K896	Data ready, text needed
S2 (Gamma conflict)	R2	K892	Data ready, text needed
S3 (Internal gamma inconsistency)	R2	K892	Will resolve with S2
S4 (TSMC gamma untraceable)	R2	K892	Will resolve with S2
S5 (VT tables no source)	R3	K900	Data ready, text needed
S6 (SSVS PIP conflict)	R3	body_v2.tex	Already correct in text

Medium Problems (8 items)

- M1** Scope creep / excessive length.
- M2** Overclaiming language (“crushes,” “fundamentally alter,” $5.0\times$ without CI in abstract).
- M3** Missing Christoffersen (1998) bibliography entry.
- M4** HAR-RV headline on native target.
- M5** GJR+Normal violation count: 9/481 vs 11/481. *Can resolve using K900 data.*
- M6** GJR QLIKE: 0.220 vs 0.217.
- M7** Integrate K892 gamma values into paper text (6+ locations).
- M8** Integrate K896 + K900 ES results into paper text.

Minor Problems (6 items)

- m1 Missing page numbers for Whaley references.
- m2 EWMA $\lambda = 0.94$ sensitivity.
- m3 Persistence formula symmetric-distribution caveat.
- m4 BPV scaling clarification.
- m5 Figure 1 vs Table 2 estimation method reconciliation.
- m6 Student-*t* df distinction.

Revision Priorities (R3, ordered)

1. **[HIGH] Integrate K900 into Tables 4–5.** Replace current mixed-period values with K900’s traceable data. Use common period (2020–2026) as the primary comparison; report full-period results as robustness. Update abstract/conclusion figures accordingly. Source attribution: “Source: K900.” *Estimated effort: 2 hours.*
2. **[HIGH] Integrate K892 gamma values (M7).** Update Table 2 (0050.TW: $\gamma = 0.097$, $t = 3.60$; TSMC: $\gamma = 0.052$, $t = 3.98$), Table 1 (summary stats), Section 3.1 (amplification narrative), Section 4.5 (TSMC now *significant*—revise discussion), Abstract, Conclusion. *Estimated effort: 1.5 hours.*
3. **[HIGH] Integrate K896 + K900 ES results (M8).** Add ES subsection to Section 7. Add Acerbi-Szekely (2014) and Fissler-Ziegel (2016) citations. Extend prediction–VaR paradox discussion to ES. Update abstract. *Estimated effort: 1.5 hours.*
4. **[MEDIUM] Fix numerical mismatches (M5, M6).** Use K900’s VaR Trinity data to update violation counts and QLIKE values. Add Christoffersen (1998) to bibliography. *Estimated effort: 30 minutes.*
5. **[MEDIUM] Tighten language (M2).** Remove “crushes” and “fundamentally alter.” Pair $5.0\times$ with CI [2.8, 8.1] in abstract. *Estimated effort: 30 minutes.*
6. **[MEDIUM] Consider restructuring (M1).** Evaluate splitting HF section into companion paper. *Estimated effort: 1 hour (decision) + 2–3 hours (if splitting).*
7. **[LOW] Fix minor issues** (Whaley page numbers, EWMA sensitivity, persistence caveat, Figure 1 reconciliation, Student- t df). *Estimated effort: 1 hour.*

Progress Assessment

Category	R1	R2	R3	Change (R2→R3)
Severe issues	6	2	0	−2 (S5 via K900, S6 clarified)
Medium issues	8	8	8	±0
Minor issues	7	6	6	±0
Paper text changes	0	0	0	No text edits yet
Experiment data ready	2	2	4	+2 (K900 for S5, S6 confirmed)

Data completeness assessment:

Paper Component	Data Source	Status
Table 1 (Summary stats)	K892	Ready
Table 2 (Gamma parameters)	K892	Ready
Table 3 (SSVS PIP)	K461, body_v2.tex	Correct in text
Table 4 (VT performance, mixed)	K900	Ready
Table 5 (VT performance, common)	K900	Ready
Section 5 (High-frequency)	K849	Already traceable
Section 6 (Macro indicators)	No JSON	Still untraceable
Section 7 (VaR)	K896, K900	Ready
Section 4.5 (TSMC concentration)	K892 (partial)	Partially traceable
Appendix (Time-zone)	No JSON	Still untraceable

Conditional recommendation: *Minor revision.* All SEVERE issues are resolved at the data level. The paper’s empirical foundation is now solid: K892 (gamma), K896 (ES), and K900 (VT performance) provide traceable, verifiable data for all core tables. The remaining work is text integration—updating 6+ tables and adding the ES section—estimated at 4–6 hours. Once the text is updated, the paper should be ready for submission to JBF or IRFA with only the 8 MEDIUM issues as “nice to have” improvements (scope tightening, language, bibliography).

Remaining risks:

- Section 6 (macro indicators) and Appendix (time-zone) still lack dedicated experiment JSONs. These are secondary contributions but should be made traceable before submission.
- The conditional leverage $+0.162$ claim needs its own traceable experiment (not in K900).
- The mixed-period Table 4 numbers will change substantially when updated with K900 data—the narrative must be adjusted accordingly (e.g., B&H Sharpe $0.729 \rightarrow 1.247$ over 2019–2026).